

Lab 12 Counting

Objective

Solving exercises from the textbook.

Current Lab Learning Outcomes (LLO)

By completion of the lab, the students should be able to

1. Understand basics of counting and pigeon hole principle
2. Solve shorter/easier or longer / harder problems given in the textbook.

Lab Requirements

Students allowed using their lecture notes in the lab and use blackboard slides in order to solve the exercises.

Lab Assessment

- 1- Divide students to groups and let them to solve the given example.
- 2- Discuss the answers with the groups and write on board the optimal solution.

Lab Description

In this lab, students learn how to apply the basic counting principles, the pigeonhole principle, and permutations and combinations to solve counting problems

1. A wired equivalent privacy (WEP) key for a wireless fidelity (WiFi) network is a string of either 10, 26, or 58 hexadecimal digits. How many different WEP keys are there?
2. Use the principle of inclusion–exclusion to find the number of positive integers less than 1,000,000 that are not divisible either by 4 or by 6.
3. How many ways are there to arrange the letters a , b , c , and d such that a is not followed immediately by b .
4. Show that among any group of five (not necessarily consecutive) integers, there are two with the same remainder when divided by 4.
5. What is the minimum number of students, each of whom comes from one of the 50 states, who must be enrolled in a university to guarantee that there are at least 100 who come from the same state?
6. Find the value of each of these quantities.
 - a. $P(6, 3)$
 - b. $P(8, 8)$
 - c. $C(5, 3)$
 - d. $C(8, 8)$
7. How many bit strings of length 10 contain
 - a) Exactly four 1's?
 - b) At most four 1's?
 - c) At least four 1's?
 - d) An equal number of 0's and 1's?
8. In how many ways can a set of five letters be selected from the English alphabet?
9. A coin is flipped 10 times where each flip comes up either heads or tails. How many possible outcomes
 - a) Are there in total?

- b) Contain exactly two heads?
 - c) Contain at most three tails.
 - d) Contain the same number of heads and tails?
10. How many permutations of the letters *ABCDEFGH* contain
- a) The string *BCD*?
 - b) The string *CFGA*?
 - c) The strings *BA* and *GF*?
 - d) The strings *ABC* and *DE*?
 - e) The strings *ABC* and *CDE*?
 - f) The strings *CBA* and *BED*?
11. How many ways are there for eight men and five women to stand in a line so that no two women stand next to each other? [*Hint*: First, position the men and then consider possible positions for the women.]
12. How many bit strings of length 10 contain at least three 1's and at least three 0's?
13. How many bit strings contain exactly eight 0's and ten 1's if every 0 must be immediately followed by a 1?
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